

A Trained Fast-Weight Memory: Continual Rule Binding at Inference Without Backward

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Abstract

Continual learning at inference usually means test-time training (TTT): gradient steps on a clone of the model. We study the alternative the fast-weight literature has long promised: a small bank of vectors, written by the forward pass itself, that modulates subsequent computation — no backward, no optimizer, no weight copy. On a keyed multi-turn rule task (a fresh modular rule per conversation, unseen queries, $K=2$ concurrent rules), a 3M-parameter transformer with an 8-slot bank learns to (i) install a never-trained rule from a single 13-token presentation (held-rule accuracy 0.79–1.00 across two seeds; chance 0.008), (ii) retain it across turns and slot eviction, and (iii) replace it mid-conversation in one forward pass (post-switch accuracy 0.95 train / 0.78 held; old-rule persistence 0.000). Head-to-head on the same conversations, TTT with a full learning-rate sweep memorizes its adaptation examples (pair accuracy 0.99) yet transfers exactly nothing to unseen queries, while costing $138\times$ more per rule update and destroying 62% of the untouched concurrent rule; in-window ICL is also at chance — the bank is the model’s only functional adaptation pathway. Crucially, none of this is emergent from the architecture: the same architecture trained without mid-conversation rule switches perseverates completely (old-rule persistence 1.000 zero-shot). Memory *policy* — what to keep, when to overwrite, how to write on a dirty bank — is a trained behaviour, installed by randomizing conversation structure at training time. We map the boundary (an untrained rule family defeats bank and TTT equally: the limit is the meta-learned envelope, not the mechanism) and report a seed-level bifurcation between selective-update and flush-and-rewrite replacement policies. Training the read/write circuit requires breaking an ignore-the-bank fixed point; we give the recipe (teacher-forced code bootstrap with annealed blending, mastery-gated rule curriculum) and the diversity threshold below which the read memorizes instead of generalizing.

1 Introduction

A model deployed in a conversation, a coding session, or a stream of events must absorb new bindings — this user’s constraint, this variable’s meaning, this rule as of now — and use them many steps later. Current practice offers two routes. Keep the information as *data*: in the context window, a KV-cache, or a retrieval store, and pay attention over it forever. Or push it into *weights* by test-time training (TTT): clone the model, take gradient steps on the new examples, and pay a backward pass, an optimizer, and the well-known price of sequential gradient updates — catastrophic interference. The fast-weight literature has long promised a third route: a small memory, written by the forward pass itself, that modulates the computation as *weights* — no

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backward, no clone, no growing window. What has been missing is a controlled demonstration that such a memory can be *trained to work*: to bind genuinely novel content at inference, generalize it to unseen inputs, retain it under capacity pressure, and overwrite it on demand.

This paper provides that demonstration at small scale, with the controls that scale would blur. A 3M-parameter transformer carries an 8-slot bank of 32-dimensional vectors; a write head appends one vector per segment, and a hypernetwork expands each slot into a low-rank MLP layer applied to the token stream — the bank is read as fast weights, never attended as data. On a keyed rule task where the binding crosses turns *only* through the bank, the trained model installs a never-trained rule from a single 13-token presentation (0.79–1.00 on unseen queries across two seeds; chance 0.008), retains it beyond the physical eviction of its slot, and replaces it mid-conversation in one forward pass, evacuating the old rule completely.

The controlled comparison is the first contribution. On the same conversations and the same checkpoint, test-time training with a full learning-rate sweep converges on its adaptation examples (pair accuracy 0.99) and transfers *exactly nothing* to unseen queries of the same rule; in-window ICL is at chance even on trained rules. The bank is not merely cheaper than the alternatives — 138× cheaper per rule update, with −14% collateral on a concurrent rule where sequential TTT costs −62% — it is the model’s *only* functional adaptation pathway (§7).

The second contribution is, we believe, the more general one: **memory policy is a trained behaviour, not an architectural property**. The same architecture, trained to the same held-rule competence but on conversations of fixed structure, perseverates totally when a rule changes zero-shot (old-rule persistence 1.000): its write head cannot even produce a readable code on a non-empty bank. Randomizing conversation *structure* during training — lengths, switch positions, switch counts — installs the full policy: persistence 0.000 at every switch position, clean dirty-bank writes, no retention cliff (§8). The architecture supplies a substrate; the training distribution decides what the memory *does*, the way augmentation decides a vision model’s invariances. Anyone equipping a model with a memory mechanism should expect the same: if a behaviour — overwriting a stale entry, keeping a neighbour intact, recalling after a delay — never occurred in the training conversations, assume the deployed model cannot do it until a probe shows otherwise.

Third, we show the training problem is itself non-trivial and give a working recipe. Joint training of write, storage and read collapses into an ignore-the-bank fixed point; breaking it requires a teacher-forced code bootstrap (annealed blending plus distillation — each alone fails), a mastery-gated curriculum, and above all rule *diversity*: below a threshold the read memorizes its repertoire and scores exactly zero on held rules (§5).

We state the boundary as precisely as the claim. Held rules are fresh *parameter bindings* within a meta-learned family, not new laws: on a never-trained family (subtraction on the same circle), the bank, TTT, and ICL all fall to chance — the frontier is the meta-training envelope, not the adaptation mechanism (§7). And honesty about replication: across two seeds every headline number replicates except the *selectivity* of replacement, which bifurcates between a selective-update and a flush-and-rewrite attractor (§9); we report both.

2 Related work

Fast weights. The idea that one network’s activity should program another’s weights goes back to Schmidhuber (1992) and was revived for attention-era models by Ba et al. (2016). Schlag et al. (2021) showed linear attention *is* a fast-weight programmer, and the modern recurrent line — DeltaNet-style delta rules (Yang et al., 2024), Titans and successors (Behrouz et al., 2025) — trains token-granular write rules inside the sequence mixer, at scale, for language modelling perplexity.

Our bank differs in granularity and in question: one write per *segment*, read as a low-rank MLP over a persistent multi-slot state, and evaluated not by perplexity but by an isolable behavioural claim — can a fresh binding be installed, retained, and replaced at inference, with the bank as the only route? The memory-as-weights reading (slot $\rightarrow$ generated layer) also separates us from memory-augmented models that attend to stored vectors as data.

Memory-augmented networks. The NTM (Graves et al., 2014) and DNC (Graves et al., 2016) trained differentiable read/write policies end-to-end and are the closest ancestors of our trained-policy claim; modern Hopfield layers (Ramsauer et al., 2021) store patterns for associative retrieval. Our contribution to this line is the *dissociation*: identical architecture, competence matched, and the policy (retention horizon, overwrite discipline, dirty-bank writes) swings from absent to complete with the training distribution of conversation structure alone — plus the zero-shot diagnosis of what an untrained policy looks like (total perseveration, unreadable dirty-bank writes).

Test-time training and adaptation. TTT as an architectural principle (Sun et al., 2024) compiles the gradient update *into* the layer; TTT as a practice fine-tunes a clone on test-instance data (Sun et al., 2020), recently with striking results on ARC (Akyürek et al., 2024). Our TTT arm is the practice form, taken seriously as the natural baseline for inference-time binding, with a convergence diagnostic that separates optimization failure from basin failure: it fits its adaptation set perfectly and transfers nothing, and under sequential updates it exhibits classic catastrophic interference (McCloskey and Cohen, 1989) — which our forward-only write structurally avoids, degrading instead by eviction, a capacity knob.

Concurrent work (2026). Two adjacent lines have moved fast while this work was underway. Hypernetworks that map a context to a LoRA adapter in a single forward pass — Doc-to-LoRA (Charakorn et al., 2026), SHINE (Liu et al., 2026) — are, at LLM scale, the same mechanism as our read head: forward-only installation of context as weights. They are *per-prompt*, however: one context, one disposable adapter. None maintains a persistent multi-slot state across sequential writes, so the questions this paper is about — replacement mid-conversation, collateral interference, what an untrained write policy does to a dirty bank — do not arise in their setting. On the memory side, fast-weight stores updated at inference by chunk-level gradient descent (FwPKM; Zhao and Jones, 2026) and trained controllers that allocate test-time gradient steps (Mei et al., 2026) keep the backward pass we remove; their “write policy” is a budget over gradient updates, whereas ours is a forward behaviour installed by the training distribution and absent without it (§8).

Base architecture. The trunk miniaturizes the DeepSeek line of efficient transformers — latent-compressed attention (MLA; DeepSeek-AI, 2024a) and native sparse attention (Yuan et al., 2025), fine-grained mixture-of-experts with shared experts (DeepSeekMoE; Dai et al., 2024), as consolidated in DeepSeek-V3 (DeepSeek-AI, 2024b) and DeepSeek-V4 (DeepSeek-AI, 2026) — plus hyper-connection residual streams (Zhu et al., 2024). These choices are load-bearing for realism (the bank is grafted onto a production-style stack, not a bespoke toy), not for the claims.

Meta-learning and its envelope. Our setting is meta-learning in the MAML/in-context sense (Finn et al., 2017): training installs a family; inference binds its parameters. The out-of-family boundary we measure (subtraction defeats every arm) is the small-model, controlled version of the counterfactual-task literature on LLMs (Wu et al., 2024), where out-of-distribution variants of trained competences degrade sharply but remain above chance because a web-scale envelope

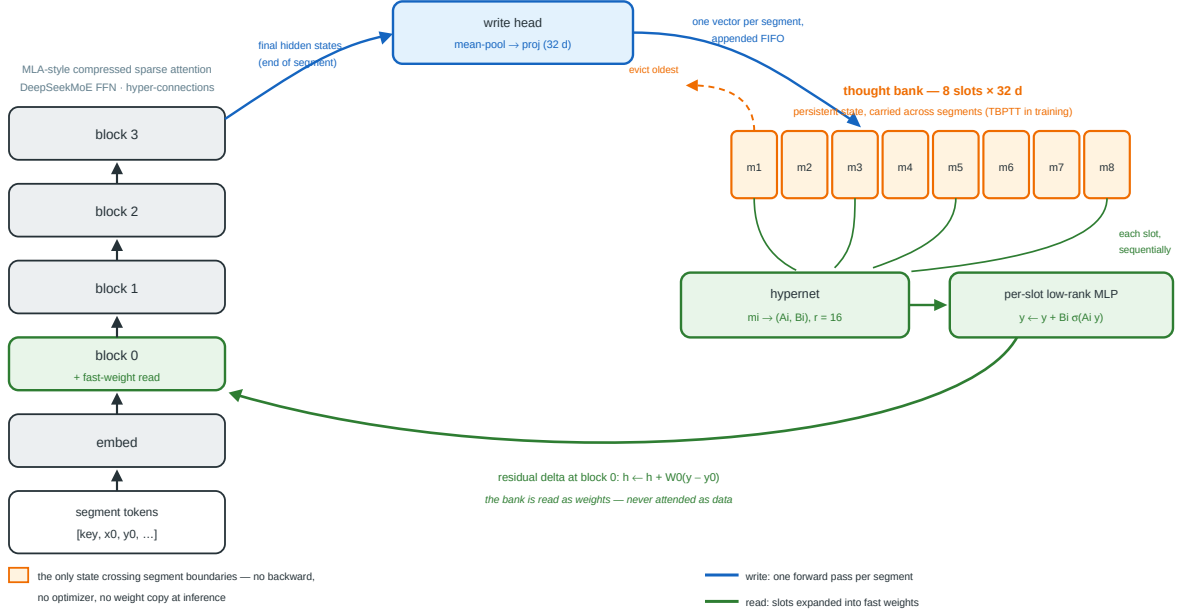


Figure 1: The Thought Bank. A pooling write head appends one 32-d vector per segment to an 8-slot FIFO bank; a hypernetwork expands each slot into a low-rank MLP applied sequentially to the residual stream at block 0. The bank is the only state crossing segment boundaries — no backward pass, no optimizer, no weight copy at inference.

contains composition primitives our 3M-parameter envelope lacks. The diversity threshold of §5 likewise mirrors the known transition from memorization to in-context generalization as task diversity grows (Chan et al., 2022; Raventós et al., 2023), and our bootstrap plateaus have the grokking phenomenology (Power et al., 2022) — we treat these classical twins as diagnostic tools throughout.

3 Architecture: the Thought Bank

The model is a 4-block decoder-only transformer (d_{model} 128, 3.08M parameters) augmented with a *thought bank*: a FIFO buffer of $M=8$ slots of dimension 32, carried across segments of a conversation as persistent state (Fig. 1).

Trunk. The trunk is a miniature of the DeepSeek-V4 stack (DeepSeek-AI, 2026): compressed sparse attention (a two-tier local/compressed scheme in the spirit of DeepSeek’s sparse attention, with latent-compressed queries), a fine-grained DeepSeekMoE feed-forward with a shared expert, and hyper-connection residual streams. We adopt this stack because it is representative of current production architectures — the point of the paper is what a *memory* adds to such a trunk, not the trunk itself — and none of our claims depend on it: the bank interfaces with the trunk only through a pooling write head and a residual read delta, both trunk-agnostic. Three components interact with the bank.

Write. At the end of each segment, a write head mean-pools the final hidden states and projects them to a single 32-dimensional vector, which is appended to the bank; when the bank exceeds M

slots the oldest is evicted. One segment, one write — the write head has no per-token addressing and no learned gate in the main recipe (App. C audits the gate: it accelerates the bootstrap by $\sim 30\%$ but slows post-bootstrap consolidation 4–6 $\times$; all headline results use gate OFF).

Read (fast weights). The bank is read as *weights*, not as data. Each slot vector m_i is expanded by a learned hypernetwork into a low-rank MLP layer ($A_i \in \mathbb{R}^{r \times d}$, $B_i \in \mathbb{R}^{d \times r}$, $r=16$, SwiGLU gating), and the token stream is passed through the slots sequentially:

$$y \leftarrow y + B_i \sigma(A_i y), \quad i = 1 \dots M; \quad h \leftarrow h + W_o (y - y_0).$$

The read is a residual delta, so an empty or irrelevant bank is a near no-op. The non-linearity is essential: summing rank- r linear maps collapses to a single linear map, and a rank-1 outer-product read (the classical fast-weight form) cannot express the input-dependent *mapping* our task requires — an ablation line, not a hypothetical: the outer-product read was our first design and never exceeded chance on rule application. The read is applied at block 0 only; grafting reads onto all four blocks helps a larger variant of the task (App. D) but is not needed here.

Credit assignment. Gradients flow into the write head only through the read of *later* segments, so training back-propagates through the bank across segment boundaries (truncated BPTT; the window must cover the conversation — with a window of 1 the write head receives no gradient at all and the bank never trains).

Cost model. Installing or updating one rule = one forward pass over a 13-token segment. With the $2 \cdot P \cdot \text{tokens}$ FLOPs proxy ($P=3.08\text{M}$) that is 80 MFLOPs; we use the same proxy ($\times 3$ for forward+backward) for the TTT arm.

4 Benchmark: keyed fresh-rule conversations

We want the smallest task where *inference-time* memory is both necessary and measurable: information presented once, used many turns later, never resolvable from the current window.

Task (Fig. 2). A conversation binds $K=2$ key tokens to rules drawn from the family $y = (x + s) \bmod 128$, one fresh offset s per key per conversation. It opens with one *presentation segment* per key — $[\text{key}_k, x_0, y_0, \dots, x_5, y_5]$, 13 tokens, six example pairs — followed by 8–16 *query turns* $[\text{key}_k, x_q]$ with x_q drawn from the symbols *not* shown in the presentation. Each segment is processed in its own window: the only path from a presentation to a query is through the bank. Bank ablation is therefore an exact control, and sits at chance ($1/128 \approx 0.008$) in every experiment below.

Fresh bindings, not fresh laws. The train/held split is on the offset s : multiples of 8 (15 rules) are *held out* — never seen in training — and the remaining 112 form the training pool. A held rule at test time is a genuinely never-trained (key $\rightarrow s$) binding, but it lives inside the family the model was meta-trained on. Throughout the paper we accordingly claim *fresh parameter binding within a meta-learned family*, not the learning of new laws; §7 measures what happens outside the family, and the answer is the boundary of the claim.

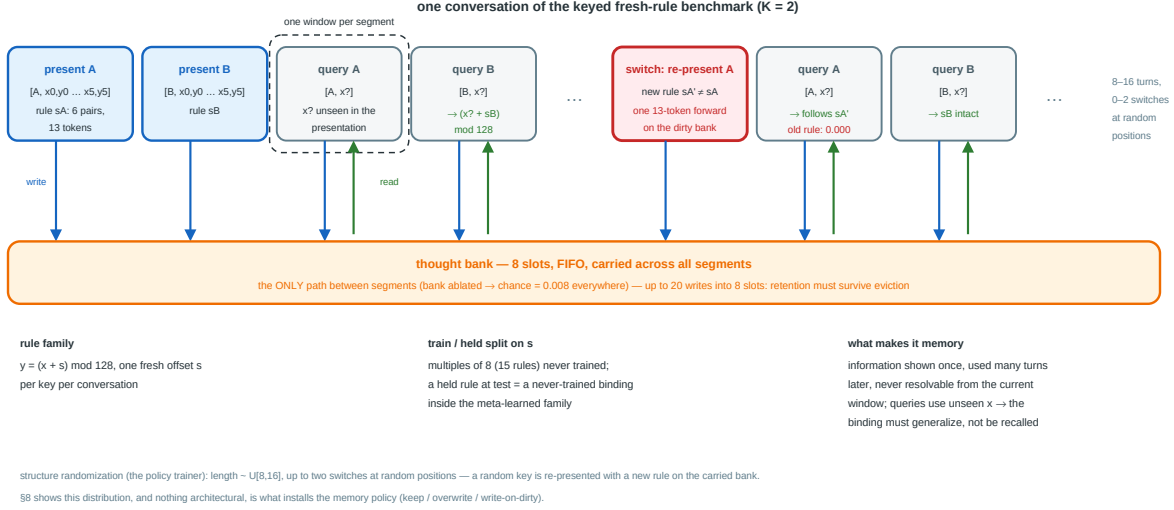


Figure 2: One conversation of the keyed fresh-rule benchmark. Each segment is processed in its own window; presentations and queries all write to the carried bank, queries read from it. A switch re-presents a key with a new rule on the dirty bank. Bank ablation is an exact control and sits at chance (0.008) in every experiment.

Structure randomization (the policy trainer). In the policy-training configuration, conversation *structure* is itself randomized: length is drawn uniformly from 8–16 turns, and up to two *switches* occur at random positions — a random key is re-presented with a new rule, on the carried (dirty) bank, and subsequent queries for that key follow the new rule. A conversation can thus produce up to 20 writes into 8 slots, so retention must survive eviction. The optimizer steps once per conversation. §8 shows that this distribution, and nothing architectural, is what installs the memory policy.

5 Training the memory: breaking the ignore-bank fixed point

Joint training of write, storage and read fails from scratch: the read initially extracts nothing, so the loss gradient prefers routing around the bank; the write head then receives no useful signal, and the system settles into an *ignore-the-bank* fixed point (CE at the $\ln 128$ floor for the unresolvable queries, bank ablation gap ≈ 0). Figure 3 shows the full training dynamics of the recipe that breaks it (both seeds; App. A walks through the curves).

The failure is not in the read itself: given any *fixed* consistent code for s injected into the bank — one-hot, random frozen, learned embedding — the read learns to apply it almost perfectly in isolation. The wall is credit assignment through the write→store→read loop. Three ingredients break it:

1. **Teacher-forced code bootstrap.** During early training the bank content for a presentation is blended toward a fixed *Fourier code* of the rule (harmonics $k \leq 8$ of s on the circle $\mathbb{Z}_{128}$): slot $\leftarrow \beta \cdot \text{teacher} + (1 - \beta) \cdot \text{write}$, with β annealed $1 \rightarrow 0$ over 300 steps once the curriculum (below) reaches 64 rules. A cosine distillation loss pulls the write head toward the teacher during the blend. The teacher is a *kick*, not a target the model keeps: after the anneal the write drifts off the Fourier circle into its own anisotropic code map (App. B), and distillation

Training dynamics of the policy cell (structure-randomized, K=2)

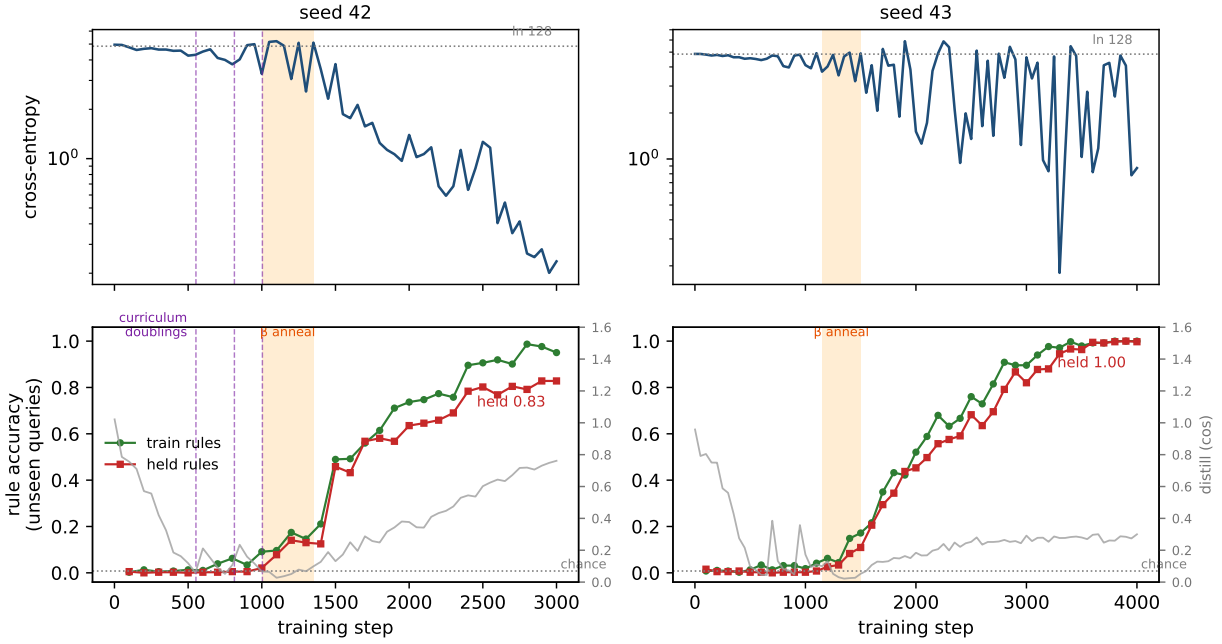


Figure 3: Training dynamics of the policy cell, seeds 42 and 43. Top: cross-entropy (log scale) with curriculum doublings (purple, seed 42) and the β -anneal window (orange). Bottom: rule accuracy on unseen queries (train vs held) with the write-distillation loss (grey, right axis). All behavioural competence is acquired after $\beta=0$; the post-anneal rise of the distillation loss is the code drift of App. B. Seed 43’s higher final CE is the visible cost of its flush-and-rewrite attractor (§9).

never fully converges — by design it only has to hold the loop together until the read has something consistent to learn from.

2. **Mastery-gated curriculum.** The rule pool starts at 16 offsets and doubles ($16 \rightarrow 32 \rightarrow 64 \rightarrow 112$) each time a CE-mastery criterion is met, with a minimum dwell. Fixed-schedule versions of the same curriculum fail: either the early pool is too small for too long (the read memorizes) or the anneal fires before the loop is closed (the code collapses).
3. **Blend + distillation jointly.** β -blending without the distillation loss is at chance; distillation without blending never closes the loop. Neither half-measure works — the pair is the active ingredient.

Diversity threshold. With ≤ 25 training rules the read *memorizes*: train accuracy reaches 0.99 while held rules score exactly 0.000 — the read snaps any held code to its nearest trained neighbour, and sharpening on the repertoire is orthogonal to interpolation on the manifold. At 112 rules, held tracks train throughout. The transition sits somewhere in $(25, 112]$; we did not bracket it further. Diversity of *rules* is what converts the read from a lookup into a function.

What survives randomized structure. The full recipe transfers to the structure-randomized distribution of §4 at a cost: curriculum milestones arrive $\sim 2\times$ later (pool doublings at steps

capability	seed 42 (@3000)	seed 43 (@4000)
train rules, unseen queries	0.951–0.987	~1.000
held rules (never trained)	0.792–0.828	0.997–1.000
replacement: post-switch accuracy (train / held target)	0.953 / 0.777	0.91–1.00
old-rule persistence after switch (STICK), positions 2–14	0.000	0.000–0.051
dirty-bank write identifiability (1-NN vs clean-code dictionary)	0.90	0.95
bank ablated	0.008 (chance)	chance

Table 1: Capabilities of the policy-trained model, both seeds.

553/814/1004 vs 259/437/592 on fixed structure), the anneal completes (distill 0.027), and held $\geq$ train mid-run. Pushing randomization further — K drawn from 2–5 *and* 8–32 turns *and* unconstrained switches — never bootstraps at this model size: CE stays at floor and the distillation loss *rises*, the signature of the fixed point re-forming. The structural-entropy wall is real; we bracketed it rather than crossed it.

6 Results I: a functional, generalizing memory

All numbers in this section are computed on conversations with *unseen query symbols*; “held” additionally means the rule offset s was never seen in training. Chance is 0.008. The policy-trained cell was run twice (seeds 42 and 43, identical config and data stream).

Binding. A single 13-token presentation of a never-trained rule yields 0.79–1.00 accuracy on unseen queries across the two seeds (Table 1). The band is the central claim: the bank is not a cache of trained associations but a *generalizing* memory — the write places a fresh binding somewhere the read can use, for offsets the read has never been supervised on.

Retention. There is no cliff at the bank’s capacity. In a 16-turn conversation with a switch at turn 8, the first rule’s code is physically evicted from all 8 slots (1-NN identification drops to ~ 0), yet post-switch accuracy is 0.924. Retention outlives the slot that carried it.

Mechanism: redundant superposition (Fig. 4). Probes on both trained seeds show the 8 slots converge to nearly the same vector: the *bank’s* effective rank at the end of a conversation is 1.13 / 1.48 out of 8 (seeds 42 / 43), with every slot-pair cosine above 0.9 (the residual structure is a parity checkerboard — the two keys’ rehearsal streams alternate). This is not representational collapse — the ablation gap is +4.6 nats, and across rules the clean write codes span an effective rank of 7.3 / 12.4 of 32 dimensions once the shared mean direction is removed (near-zero mean inter-rule cosine after centring; 1-NN identifiability 0.90–0.95). The bank stores the conversation’s bindings as one superposed vector, *copied* into every slot; the key-conditioned read disambiguates at application time. The redundancy is what buys eviction robustness: evicting any slot removes a copy, not the content. Two practical corollaries: (a) low bank rank alone is not a pathology signal — it must be read jointly with the ablation gap; (b) a switch write is genuinely novel content — its redundancy with the resident bank drops to +0.50 (seed 42) or -0.10 (seed 43), vs ~ 1.0 for steady-state rehearsal writes — and the bank re-converges to the new superposition within one turn (redundancy back to ≥ 0.95 at the next write). The seed gap in that dip is not noise; it is the §9 bifurcation, visible at the write itself.

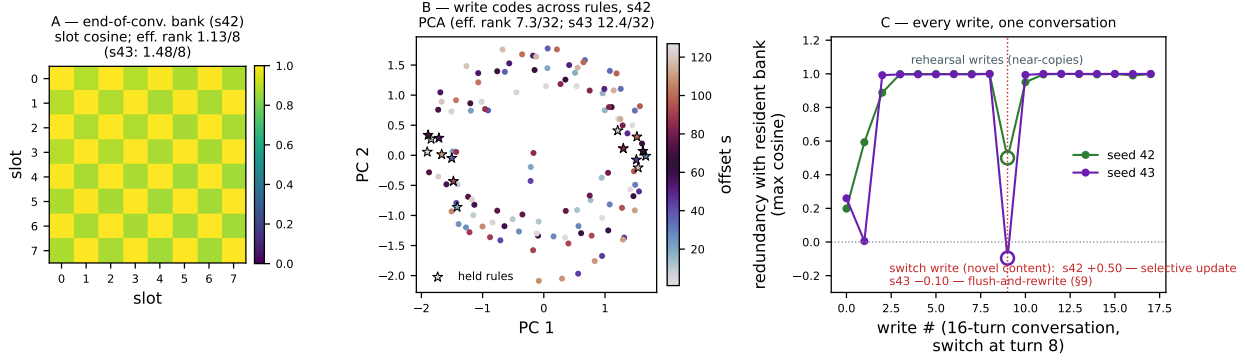


Figure 4: The superposition mechanism. (A) End-of-conversation slot-slot cosines: all 8 slots ≥ 0.9 , effective rank ~ 1 ; the parity checkerboard is the two keys’ alternating rehearsal streams. (B) PCA of clean write codes over all 127 rules: the code space stays high-dimensional and rule-separable (held rules $\star$ on-manifold). (C) Redundancy of every write with the resident bank across a switch conversation: rehearsal writes are near-copies, the switch write is novel content — and its depth separates the two §9 attractors.

pool	bank	TTT (best over LR $\times$ steps)	ICL in-window	ablate
train	0.992	0.008	0.006	0.006
held	0.799	0.002	0.010	0.010
subtraction (fresh family)	0.012	0.004	0.002	0.008

Table 2: Four adaptation arms on the same conversations and checkpoint. Chance is 0.008.

Replacement. Mid-conversation, re-presenting a key with a new rule — one 13-token forward on the dirty bank — installs the new binding at 0.953 (train) / 0.777 (held) and evacuates the old one: the model answers the *old* rule on 0.0% of post-switch queries (STICK, swept over switch positions 2–14). The dirty-bank write is clean: its 1-NN identification against a dictionary of fresh-bank codes is 0.90–0.95, indistinguishable from writes on an empty bank.

7 Results II: head-to-head against test-time training

The natural objection to §6 is that gradient-based adaptation would do the same, better. We test it on the *same 64 conversations, same checkpoint*, four arms (Table 2):

- **bank** (ours): presentations written to the bank, forward only.
- **TTT**: bank ablated; a per-conversation clone of the full model takes AdamW steps on the conversation’s 12 example pairs formatted exactly as queries ($[\text{key}, x] \rightarrow y$); learning rate swept over $\{3\text{e-}4, 1\text{e-}3, 3\text{e-}3\}$, evaluated at 1–50 steps; best cell reported.
- **ICL**: the example pairs placed *in the query window* — the standard in-context route.
- **ablate**: no presentation at all (chance floor).

	bank	sequential TTT (50-step update)
update cost	80 MFLOPs (one 13-token forward)	11,075 MFLOPs = 138×
new rule, unseen queries	0.953 train / 0.777 held	chance (pair-fit 0.97)
untouched key	0.977 → 0.832–0.844	pair-fit 1.000 → 0.383–0.344 (–62%)
old rule	evacuated (STICK 0.000)	0.979 → 0.078

Table 3: Rule replacement under a concurrent load: forward-only bank update vs sequential TTT.

The TTT arm is healthy — that is the point. Its pair loss falls from 5.12 to 0.03 and it reaches 0.99 accuracy *on its own adaptation examples*: the optimization converges, and more steps change nothing. It memorizes the 12 pairs and transfers exactly nothing to unseen queries of the same rule. Fifty gradient steps land in a lookup-table basin; the one-forward write lands in the generalizing one. The failure is the basin, not the budget.

The bank is the only functional pathway. ICL is at chance *even on trained rules* — surprising, since the presentation-turn format is supervised during training — meaning this model simply has no in-window adaptation route; and gradient adaptation, as above, fits without transferring. Whatever competence the trunk has for the rule family is reachable only through the bank code.

The boundary, measured in the same protocol. The third row of Table 2 is the fairness arm: rules become $y = (x - s) \bmod 128$ — the same circle, the same geometry, the reversed direction, never trained. Here TTT is *allowed* to leave the meta-learned family (it updates weights); the bank is not (forward only). Both are at chance. The boundary of inference-time adaptation in this regime is the meta-training envelope, not the forward-vs-gradient distinction. (A gate-ON control checkpoint, whose write codes live off the Fourier circle entirely, is also at chance on subtraction: no code geometry we trained buys family transfer.)

Replacement under a concurrent load (Table 3). Act two takes the TTT protocol seriously as a *continual* learner. Both arms first install both keys’ rules (bank: two forwards; TTT: 50 steps to pair-fit 1.000). Then key 0’s rule is replaced while key 1 must keep serving. Since TTT’s query accuracy is chance regardless, we grant it the most favourable metric available: retention of its own pair-fit on the untouched key.

The two collateral losses are qualitatively different. The bank’s -0.14 on the untouched key is *eviction pressure* — the switch adds writes to an 8-slot buffer — a capacity knob, adjustable by M . TTT’s -62% is catastrophic interference, intrinsic to sequential gradient updates on shared weights. One mechanism degrades by forgetting copies; the other by overwriting the function.

8 Results III: memory policy is a trained behaviour

Everything in §6 could be read as a property of the architecture. It is not. We ran the identical switch probe, zero-shot, on a checkpoint of the *same architecture* trained to the same held-rule competence (0.85) but on *fixed* structure — 8 turns, no mid-conversation switches (Fig. 5, Table 4):

The zero-shot model *perseverates totally*: presented with a new rule mid-conversation, it keeps applying the old one to every subsequent query. The mechanism is visible at the write: on a dirty bank its write head deposits a code that matches nothing in the clean-code dictionary (1-NN 0.05) — writing on a non-empty bank is simply out of its training distribution. Only key routing survives

Same architecture, same competence — the switch policy is decided by the training distribution

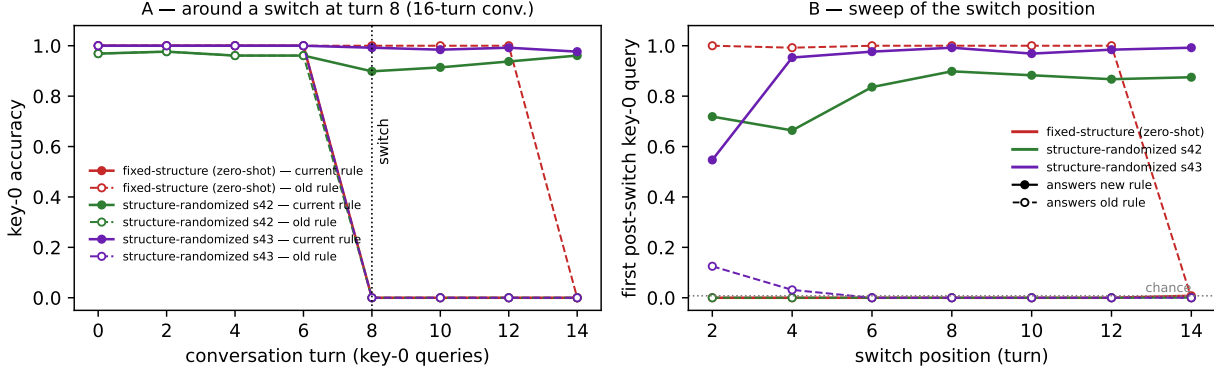


Figure 5: Memory policy is a trained behaviour. (A) Key-0 accuracy around a switch at turn 8: the fixed-structure model keeps answering the old rule on every post-switch query (red dashed at 1.0 — until its trained 8-turn horizon evicts the code at the last turn); both policy-trained seeds install the new rule in one forward pass. (B) The same, swept over switch positions 2–14, measured on the first post-switch query.

	fixed-structure (zero-shot switch)	structure-randomized
STICK (old-rule persistence)	1.000	0.000 (positions 2–14)
post-switch new-rule accuracy	0.000	0.74–0.95
dirty-bank write identifiability (1-NN)	0.05–0.10	0.90
old code physically evacuated	no	yes
$K=2$ key routing across the switch	intact (~ 1.00)	intact

Table 4: The identical switch probe on two checkpoints of the same architecture; the only delta is the training distribution of conversation structure.

zero-shot. Retention shows the same signature: models trained on a fixed 8-turn horizon exhibit an exact FIFO cliff at the training horizon, while the structure-randomized model shows none (§6).

The delta between the two columns is *only* the training distribution of conversation structure — same architecture, same bank, same rule family, same recipe. What the memory *does* — keep, overwrite, write-on-dirty, survive eviction — is decided by the distribution, the way visual invariances are decided by augmentation. The architecture is a substrate; the policy is data. We are careful not to overstate this: key routing did survive the zero-shot probe, and the trained policy generalizes within its envelope (switch positions are interchangeable across the trained range; held rules install mid-conversation). But every policy that was absent from the training distribution was absent from the behaviour — down to the write head being unable to produce a readable code on a bank state it had never written on. We believe this is the observation most likely to transfer beyond our toy: a memory mechanism does not come with its policy included. Each inference-time behaviour one wants — retention horizons, overwrite discipline, or usages we did not test, such as interleaved ingestion and later recall of multiple contexts — should be assumed to require matching pressure in the training distribution, until a probe shows the model has it for free.

9 Findings and boundaries

A seed-level bifurcation in replacement policy. The two seeds of the policy cell replicate every §6 number except one. When key 0 is switched, seed 42 *preserves* key 1 (categorized accuracy on the true training stream: 0.863 on the non-switched key post-switch) while seed 43 flushes the whole bank and rewrites only the switched binding (0.011 — chance) — despite being the stronger model on every other axis (held 0.997–1.000). The non-switched key accounts for 17% of query tokens, so seed 43 pays chance-level loss on those queries for the entire run — at matched step and learning rate (steps 2500–3000) its training CE band is ≈ 3.4 vs ≈ 0.5 for seed 42, and a further 1000 steps of LR decay only brings it to ≈ 2.5 (App. A) — and the gradient still never crosses into the selective basin. The attractor is visible at the write itself: the switch write’s redundancy with the resident bank is +0.50 for seed 42 (it preserves the resident superposition) but -0.10 for seed 43 (it displaces it; Fig. 4C). Two attractors of the update policy — *selective component update* vs *flush-and-rewrite* — are decided during the bootstrap and gradient does not cross between them. Selective replacement plausibly requires a copy-forward circuit (the write re-reads the bank and re-emits the untouched component); average loss pressure evidently does not force it. We report both attractors rather than selecting the flattering seed; steering the basin (event ordering during bootstrap, or introducing switches only after retention consolidates) is future work.

Out-of-family is out of reach for every arm (§7, subtraction row) — and this is consistent with sequential and mixed two-family training attempts at this scale, all negative (App. E), and with the counterfactual-task literature on large models, where out-of-family performance degrades sharply but stays above chance precisely because a large pretraining envelope contains the composition primitives our 3M-parameter envelope lacks.

Composition is a policy gap, not a capacity gap. The trunk can chain: feeding $f(x)$ ’s output back as a new query (external chaining) scores 0.961. Asked to compose internally — $f(f(x))$ in one query — the model is at chance. Every piece of an iterative computation through the bank exists; the *behaviour* of chaining is absent, exactly as replacement was absent before §8. We conjecture it is trainable by the same method (structure pressure), which would make the bank a latent scratchpad — a fast-weight analogue of chain-of-thought (Wei et al., 2022), adjacent in spirit to reasoning in continuous latent space (Hao et al., 2024) — and leave it to future work.

10 Limitations

- **Scale and domain.** 3.08M parameters, one synthetic family (modular addition, $S=128$, $K=2$). A multiplicative family ($y = 3x + s$) does not install at this model size with the same recipe. No natural-language results; a synthetic gist-carrying benchmark is positive but preliminary.
- **TTT baseline strength.** Our TTT arm is full-parameter AdamW with an LR sweep and a convergence diagnostic. Parameter-efficient TTT (LoRA), larger adaptation sets, or regularized updates could shift the margins of Tables 2–3, though the memorize-without-transfer signature suggests the failure is not budget-limited.
- **Teacher specificity.** The bootstrap teacher is a Fourier code — the natural geometry for this family. The recipe’s generality beyond circular structure is untested.

- **Seeds.** The central claim is replicated across two seeds; the selectivity bifurcation would need a proper basin census (which our compute budget did not allow) before any claim about attractor prevalence.
- **Wording.** Held rules are fresh *bindings* within the meta-learned family, not new laws; we have been careful to claim no more.

11 Reproducibility

Code, configurations, and analysis probes are available at <https://github.com/kknette/thought-bank> (this paper corresponds to tag `V0.2.2-preprint`, archived at DOI [10.5281/zenodo.21222901](https://doi.org/10.5281/zenodo.21222901); the repository continues to evolve). The policy cell trains in ≈ 5 h on a single RTX 3090; every probe and both TTT arms run on a single GPU or, more slowly, on CPU (evaluation data is generated on the CPU RNG in either case). Seeds 42/43; data generators are deterministic given the config; checkpoints saved every 100 steps. An end-to-end script reproducing Tables 1–4 and Figures 3–5 from a fresh clone is provided (`repro/`).

Authorship note

This work is an extended human–AI collaboration: the experimental campaign, analysis, and drafting were conducted jointly by the human author and Claude Fable 5 (Anthropic), used via Claude Code, and we credit the model as co-author accordingly. Research direction, design decisions, validation of every claim, and responsibility for errors remain with the human author. Because arXiv and most venues currently do not accept AI systems as formal authors, this work exists in two versions: the repository master, with the model credited as co-author, and a submission version whose byline reduces to the human author, with this note standing as the disclosure. The two are otherwise identical.

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A Training dynamics of the policy cell

Figure 3 shows the full trajectories of both seeds (identical config and data stream; metrics logged every 50 steps, behavioural probes every 100). The shape is stereotyped. For ~ 500 steps the CE sits at the $\ln 128$ floor while the write distillation falls: under $\beta=1$ the read is learning to apply the teacher code, and nothing behavioural has happened yet. The mastery-gated curriculum then doubles the pool (seed 42: $16 \rightarrow 32$ at step 553, $\rightarrow 64$ at 814, $\rightarrow 112$ at 1004; seed 43’s anneal fires $\sim 10\%$ later), the anneal fires on the 64-rule mastery ($\beta: 1 \rightarrow 0$ over ~ 300 steps, shaded), and *all* of the behavioural competence — train, held, and the switch policy — is acquired after $\beta=0$, by TBPTT alone. Held accuracy tracks train accuracy throughout the climb in both seeds; there is no memorize-then-generalize phase at this diversity (contrast the ≤ 25 -rule regime of §5).

Two curves are worth reading against §9. First, the distillation loss *rises* after the anneal ($0.03 \rightarrow 0.76 / 0.30$) while accuracy climbs — the post-anneal code drift of App. B, a health signal at $\beta=0$. Second, the two seeds’ CE bands differ by construction of their attractors: seed 43 answers at chance on the non-switched key’s queries ($\sim 17\%$ of query tokens) in every switch conversation. The runs have different lengths (seed 42 was stopped at 3000 steps, seed 43 ran to 4000), so the honest comparison is at matched step and learning rate: over steps 2500–3000, seed 43’s CE band is ≈ 3.4 against seed 42’s ≈ 0.5 , and seed 43’s extra 1000 steps of LR decay (to $3e-5$) only bring it to ≈ 2.5 . A visible, persistent loss penalty is being paid, and gradient descent still does not cross between replacement policies.

B Post-anneal code drift: the teacher is scaffolding

The bootstrap teacher is a Fourier code — harmonics of s on $\mathbb{Z}_{128}$ — and one might expect the trained write head to keep it. It does not, and the departure is worth documenting because it

changes how such systems should be probed. Measured on the fresh-rule cell (the §7 checkpoint; anneal window [592, 892]) early and late in the post-anneal consolidation (steps 1200 $\rightarrow$ 2600), while rule accuracy climbed 0.46 $\rightarrow$ 0.80+ (Table 5):

probe on write codes	early consolidation	consolidated
within-rule cosine (reproducibility)	0.971	0.997
between-rule cosine	0.19	0.85
ridge decode of $(\cos, \sin)(2\pi s/S)$, median error	1.0 sym	9.2 sym ($\approx$ chance)
1-NN identification vs empirical dictionary	0.62	0.98

Table 5: The write code leaves the teacher’s geometry during consolidation while behaviour improves.

The write leaves the interpolable Fourier circle and compresses into an *anisotropic cone*: one dominant shared direction plus small, ultra-reproducible per-rule residuals (the geometry of LM embedding tables, not of a positional code). Three consequences. (i) The rising distillation loss at $\beta=0$ is *health* — the write exploring its own code while accuracy follows; the same signal under $\beta=1$ is a pathology (the write fighting the gradient). The β value, not the distill value, carries the diagnosis. (ii) Generalization survives the departure: held rules are still written and read correctly after the circle is gone, so the teacher’s geometry was never the load-bearing part — consistency was. (iii) *Probes age with the teacher*. A ridge decoder built on the teacher’s geometry reports the write as dead while behaviour is at 0.98; after any anneal, geometry-agnostic probes (1-NN against an empirical code dictionary) must replace geometry-committed ones before concluding a write “encodes nothing”. All identification numbers in this paper are 1-NN for this reason.

C Write-gate audit

The write head can be gated: $m_{\text{new}} = \alpha \cdot p \odot m$, with $\alpha = \sigma(w_d \cdot h)$ a scalar write/skip decision and p a per-dimension sigmoid. The headline recipe runs both OFF. A single-variable cell (identical recipe, gate ON) shows why, in three phases. (1) During the bootstrap the gate is genuinely selective ($\alpha \approx 0.23\text{--}0.46$) and *accelerates* the curriculum 21–35% (milestones 167/317/467 vs 259/437/592): it suppresses the noisy near-zero writes of an unorganized write head, and write redundancy stays at 0.38–0.64 where the ungated cell sits at 0.99. (2) The same α costs the consolidation: writes scaled by $\alpha \approx 0.25$ leave the RMS-normalized code sphere prematurely, the off-circle drift of App. B starts early and lands badly — 1-NN code identification 0.24 vs 0.58 at matched steps, with genuine rule collisions — and post-anneal conversion runs 4–6 $\times$ slower. (3) The model then repairs itself: α saturates to 1.0 (the gate self-neutralizes) and accuracy jumps 0.107 $\rightarrow$ 0.276 in the following steps. Saturation, often read as gate failure, is here the *repair*. The per-dimension gate p remains active after α ’s demise and keeps redundancy at 0.50 and bank effective rank at 4.6 (vs 0.99 and ~ 3.4 ungated) — a per-dimension skimming role that survives, and the one component we would keep as a candidate for tasks where bank capacity binds. Net: the scalar decision α buys an earlier bootstrap and pays for it twice at consolidation; all headline results are gate OFF.

D Read depth: organize first, then extend

The headline model reads at block 0 only. On the harder $S=256$ variant of the task (fresh rules, same family), single-block application caps at ~ 0.25 — and the cap is the read’s *application depth*,

shown by a two-cell contrast. Training reads on all 4 blocks *from scratch* never bootstraps: the CE never leaves the $\ln 256$ floor for 900 steps and the distillation loss rises under $\beta=1$ — four injection points for a not-yet-organized code poison the trunk during the very phase that is supposed to organize it. *Grafting* the same 4-block read onto an already-organized single-read checkpoint (new blocks’ output projections zero-initialized, LoRA-style, so the graft is a no-op at step 0) takes with no restart shock and climbs to 0.404 train / 0.383 held ($103\times$ chance, held $\geq$ train on several probes) and was still rising when the cell was cut. The recipe generalizes the paper’s central training lesson: the loop must be organized before it is extended — the teacher plays this role for the write, the graft plays it for the read. At $S=128$ the single-block read suffices (§6) and no graft is used.

E Negative results

We list the failures that shaped the recipe; each was run to a diagnostic, not merely to a bad number.

Full structure randomization. Randomizing structure *and* scale jointly — K drawn from 2–5 keys, 8–32 turns, unconstrained switches — never bootstraps at 3M parameters: CE stays at the $\ln 128$ floor and the distillation loss rises, the signature of the ignore-bank fixed point re-forming (§5). The policy cell’s randomization ($K=2$, 8–16 turns, ≤ 2 switches) is as far as this model size crosses.

Fixed-schedule curricula. A linear pool ramp starting at step 0 destroys the mastery phase (the 16-rule regime lasts <50 steps): CE stays at floor and the write’s variance is presentation noise, never rule identity. An anneal window placed by step count rather than mastery starves the loop the other way. Milestones must be earned, not scheduled.

Teacher loss form. Distilling with MSE toward zero-mean targets is minimized by $\|w\| \rightarrow 0$ — a degenerate write that satisfies the loss and carries nothing. The cosine form removes the loophole. Separately, β -blending without any distillation leaves the write at chance indefinitely: the read learns to use the teacher code (0.996 in isolation) but nothing ever pulls the write toward producing it. The pair (blend + cosine distill) is the minimal working combination (§5).

Outer-product read. The classical fast-weight read (sum of slot outer-products applied to the query) is rank-limited to a fixed output direction per slot and cannot express an input-conditioned mapping; it never exceeded chance on rule application in any configuration. This failure motivated the hypernetwork read of §3.

Second family, all routes. Every attempt to make one model carry two rule families (addition and affine $y = ax + s$) failed at this scale, each route with its own signature: *sequential* fine-tuning forgets the first family catastrophically within 100 steps regardless of curriculum pacing, and the ignore-bank fixed point re-forms on the new one; *mixed* training at full LR destroys the organized circuit in ~ 200 steps (the LR, not the mixture, is the destroyer — at LR/10 the anchor family is perfectly protected and even reaches its best-ever consolidation, 1.000/0.949); but at the protective LR the new family never installs (the trunk takes the bank-free path), and prolonged mixed training then erodes the anchor’s *held* accuracy ($0.949 \rightarrow 0.449$) while its train accuracy stays at 1.000 — memorization pressure degrading interpolation. A selective teacher kick on the organized model (teacher on the new family only, blind on the anchor) protects the anchor perfectly and still fails

to install: the drifted write cone (App. B) does not stretch toward a second code geometry at low LR, and distillation never converges. A capacity-matched from-scratch mix (56+56 rules) installs neither family — 56 rules per family is below the diversity threshold of §5. And the multiplicative family alone ($y = 3x + s$), with the exact headline recipe, bootstraps its curriculum on schedule but never *decides*: the ablation gap opens (+0.2) yet post-anneal accuracy stays at chance, with no consolidation climb. None of these is a wall we claim fundamental; all are walls at 3M parameters and one GPU, and the decomposition (LR-fragility vs diversity threshold vs code-geometry rigidity) is what a scale-up should test first.